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MTH 211

Probability and Statistics

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Probability and Statistics Report

Signal Processing (Probability, Statistics and Noise)

Submitted to:

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1. Introduction:

Digital signal processing (DSP) is the process of analyzing and modifying a signal to optimize or improve its efficiency or performance. It involves applying various mathematical and computational algorithms to analog and digital signals to produce a signal that's of higher quality than the original signal.

DSP is primarily used to detect errors, and to filter and compress analog signals in transit. It is a type of signal processing performed through a digital signal processor or a similarly capable device that can execute DSP specific processing algorithms. Typically, DSP first converts an analog signal into a digital signal and then applies signal processing techniques and algorithms. For example, when performed on audio signals, DSP helps reduce noise and distortion. Some of the applications of DSP include audio signal processing, digital image processing, speech recognition, biomedicine and more.

Probability and statistics are two branches of mathematics concerning the collection, analysis, interpretation, and display of data in the context of random events. They are often studied together due to their interrelationship.

Statistics is used for data mining, speech recognition, vision and image analysis, data compression, artificial intelligence, and network and traffic modeling. A statistical background is essential for understanding algorithms and statistical properties that form the backbone of computer science. Probability is everywhere in computer science. In networks and systems, it is a key tool that allows us to predict performance, to understand how delay changes with the system parameters, and more. In algorithms, randomization is used to design faster and simpler algorithms than their deterministic counterparts.

In digital signal processing, statistics and probability are employed to describe signals and the methods that produce them. For instance, reducing interference, noise, and other unwanted elements in obtained data is one of the main uses of DSP. These could be unavoidable by products of some DSP operations, or they could be introduced as a defect in the data gathering equipment or an intrinsic component of the signal being monitored. These disruptive traits can be quantified and categorized using statistics and probability, which is the first step in creating removal techniques. The most crucial statistics and probability ideas are introduced in this chapter, with a focus on how they relate to acquired signals.

2. Analysis:

2.1. Definition of Digital Signal Processing:

Digital Signal Processors (DSP) take real-world signals like voice, audio, video, temperature, pressure, or position that have been digitized and then mathematically manipulate them. A DSP is designed for performing mathematical functions like "add", "subtract", "multiply" and "divide" very quickly.

Signals need to be processed so that the information that they contain can be displayed, analyzed, or converted to another type of signal that may be of use. In the real-world, analog products detect signals such as sound, light, temperature or pressure and manipulate them. Converters such as an Analog-to-Digital converter then take the real-world signal and turn it into the digital format of 1's and 0's. From here, the DSP takes over by capturing the digitized information and processing it. It then feeds the digitized information back for use in the real world. It does this in one of two ways, either digitally or in an analog format by going through a Digital-to-Analog converter. All of this occurs at very high speeds.

A DSP's information can be used by a computer to control such things as security, telephone, home theater systems, and video compression. Signals may be compressed so that they can be transmitted quickly and more efficiently from one place to another (e.g., teleconferencing can transmit speech and video via telephone lines). Signals may also be enhanced or manipulated to improve their quality or provide information that is not sensed by humans (e.g., echo cancellation for cell phones or computer-enhanced medical images). Although real-world signals can be processed in their analog form, processing signals digitally provides the advantages of high speed and accuracy.

2.2.Signal and Graph Terminology:

A signal is a description of how one parameter is related to another parameter. Signals formed from parameters that are quantized in this manner are said to be discrete signals or digitized signals. For the most part, continuous signals exist in nature, while discrete signals exist inside computers.

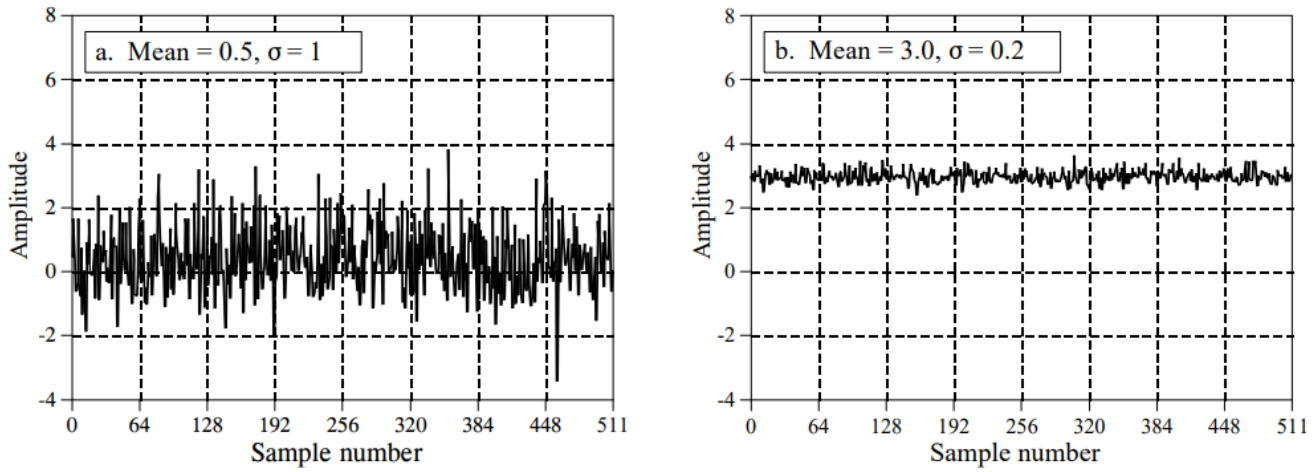


Figure 1 Examples of two digitized signals with different means and standard deviations.

The figure above shows two discrete signals, such as might be acquired with a digital data acquisition system. The vertical axis may represent voltage, light intensity, sound pressure, or an infinite number of other parameters. Since we don't know what it represents in this particular case, we will give it the generic label: amplitude. This parameter is also called several other names: the yaxis, the dependent variable, the range, and the ordinate. The horizontal axis represents the other parameter of the signal, going by such names as: the x-axis, the independent variable, the domain, and the abscissa.

2.3. Mean and Standard Deviation:

The mean, indicated by μ (a lower-case Greek mu), is the statistician's jargon for the average value of a signal. It is found just as you would expect add all of the samples together, and divide by N. It looks like this in mathematical form.

In electronics, the mean is commonly called the DC (direct current) value. Likewise, AC (alternating current) refers to how the signal fluctuates around the mean value. If the signal is a simple repetitive waveform, such as a sine or square wave, its excursions can be described by its peak-to-peak amplitude. Unfortunately, most acquired signals do not show a well-defined peak-to-peak value, but have a random nature, such as the signals

$$\mu = \frac{1}{N}\sum_{i=0}^{N-1} x_i$$

in Fig. (1) above. A more generalized method must be used in these cases, called the standard deviation, denoted by σ (a lower-case Greek sigma).

A standard deviation (or σ) is a measure of how dispersed the data is in relation to the mean. Low standard deviation means data are clustered around the mean, and high standard deviation indicates data are more spread out. It's given by the rule:

$$\sigma^2 = \frac{1}{N-1}\sum_{i=0}^{N-1}(x_i - \mu)^2$$

Figure (2) shows the relationship between the standard deviation and the peak-to-peak value of several common waveforms.

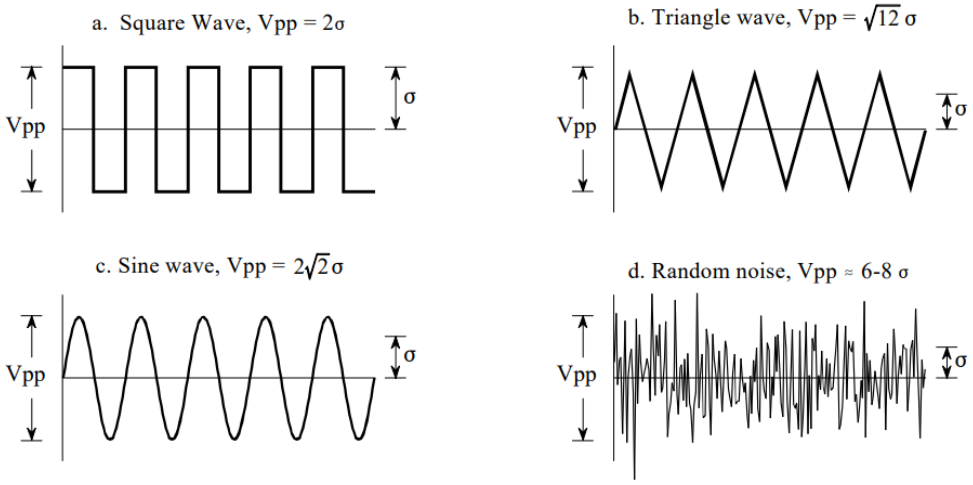


Figure 2 Ratio of the peak-to-peak amplitude to the standard deviation for several common waveforms

In some cases, the standard deviation represents noise and other interference, whereas the mean defines what is being measured. In these situations, the standard deviation only matters in respect to the mean and not by itself. The phrase "signal-to-noise ratio" (SNR), which is defined as the mean divided by the standard deviation, comes from this.The coefficient of variation is yet another expression that is employed (CV). This is calculated by multiplying the standard deviation by the mean by 100%. A signal (or any set of measurement data), for instance, with a CV of 2% has an SNR of 50. Better data means a higher value for the SNR and a lower value for the CV.

2.4. Signal vs. Underlying Process:

The study of understanding numerical data, such as captured signals, is known as statistics. In contrast, DSP uses probability to comprehend the signal-generation processes. Although they are closely linked, many DSP techniques depend on the ability to distinguish between the collected signal and the underlying process.



Figure 3 Filliping a coin

For example, imagine creating a 1000-point signal by flipping a coin 1000 times If the coin flip is heads, the corresponding sample is made a value of one. On tails, the sample is set to zero. The process that created this signal has a mean of exactly 0.5. The actual 1000-point signal will not necessarily have a mean of exactly 0.5. Random chance will make the number of ones and zeros slightly different each time the signal is generated. This random irregularity found in actual data is called by such names as: statistical variation, statistical fluctuation, and statistical noise.

2.5. The Histogram, PMF, PDF, and Normal Distribution:

The **histogram** displays the number of samples there are in the signal that have each of these possible values. The sum of all values in the histogram is equal to the number of points in the signal. The histogram is what is formed from an acquired signal. The corresponding curve for the underlying process is called **the probability mass function (pmf)**. A histogram is always calculated using a finite number of samples, while the pmf is what would be obtained with an infinite number of samples. The pmf can be estimated (inferred) from the histogram, or it may be deduced by some mathematical technique, such as in the coin flipping example. The pmf is important because it describes the probability that a certain value will be generated.

The histogram and pmf can only be used with discrete data, such as a digitized signal residing in a computer. A similar concept applies to continuous signals, such as voltages appearing in analog electronics. The **probability density function (pdf),** also called the probability distribution function, is to continuous signals what the probability mass function is to discrete signals

Signals formed from random processes usually have a bell-shaped pdf. This is called a **normal distribution**, a Gauss distribution, or a Gaussian.

2.6. Digital Noise Generation:

Both electronics and DSP have a lot to discuss about random noise. It regulates things like the smallest signal a device can measure, the maximum distance a radio system can communicate, and the amount of energy needed to create an x-ray image. Creating signals that resemble various kinds of random noise is a typical requirement in DSP. This is necessary to evaluate how well algorithms perform when there is noise present.

The heart of digital noise generation is the random number generator. Most programming languages have this as a standard function. The BASIC statement: X = RND, loads the variable, X, with a new random number each time the command is encountered. Each random number has a value between zero and one, with an equal probability of being anywhere between these two extremes.

Figure (4) shows a signal formed by taking 128 samples from this type of random number generator. The mean of the underlying process that generated this signal is 0.5, the standard deviation is $\frac{1}{\sqrt{12}}$ and the distribution is uniform between zero and one.

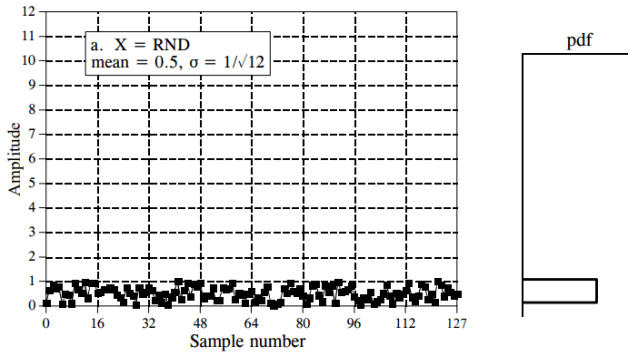


Figure 4 signal formed by taking 128 samples from this type of random number generator

Algorithms need to be tested using the same kind of data they will encounter in actual operation. This creates the need to generate digital noise with a Gaussian pdf. There are two methods for generating such signals using a random number generator.

Figure (5) shows a signal obtained by adding two random numbers to form each sample, i.e., X = RND+RND. Since each of the random numbers can run from zero to one, the sum can run from zero to two. The mean is now one, and the standard deviation is $\frac{1}{\sqrt{6}}$ (remember, when independent random signals are added, the variances also add). As shown, the pdf has changed from a uniform distribution to a triangular distribution. That is, the signal spends more of its

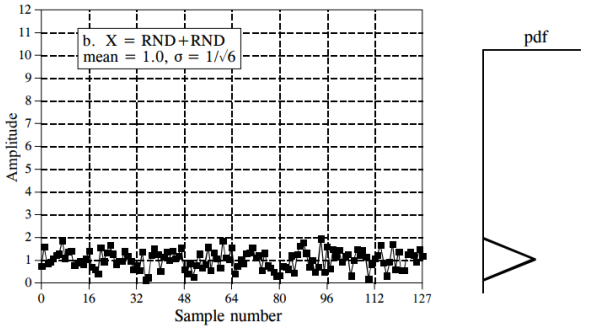


Figure 5 shows a signal obtained by adding two random numbers

time around a value of one, with less time spent near zero or two.

Figure (6) takes this idea a step further by adding twelve random numbers to produce each sample. The mean is now six, and the standard deviation is one. What is most important, the pdf has virtually become a Gaussian. This procedure can be used to create a normally distributed noise signal with an arbitrary mean and standard deviation.

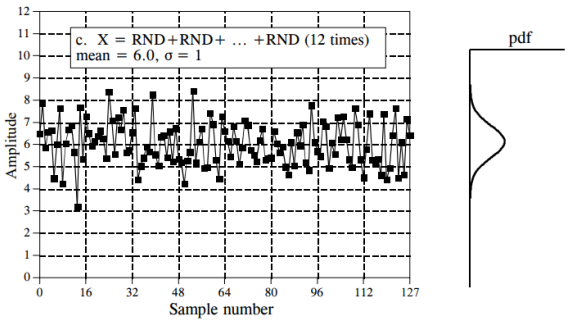


Figure 6 adding twelve random numbers to produce each sample

This procedure can be used to create a normally distributed noise signal with an arbitrary mean and standard deviation.

2.7. Accuracy and Precision:

Accuracy measures how close results are to the true or known value. Precision, on the other hand, measures how close results are to one another. Probability is useful in detecting the Precision and accuracy.

As an example, consider an oceanographer measuring water depth using a sonar system. As with all empirical measurements, a certain amount of error exists between the measured and true values. This measurement could be affected by many factors: random noise in the electronics, waves on the ocean surface, plant growth on the ocean floor, variations in the water temperature causing the sound velocity to change, etc.

To investigate these effects, the oceanographer takes many successive readings at a location known to be exactly 1000 meters deep (the true value). These measurements are then arranged as the histogram shown in Fig (7). As would be expected from the Central Limit Theorem, the acquired data are normally distributed. The mean occurs at the center of the distribution and represents the best estimate of the depth based on all of the measured data. The standard deviation defines the width of the distribution, describing how much variation occurs between successive measurements.

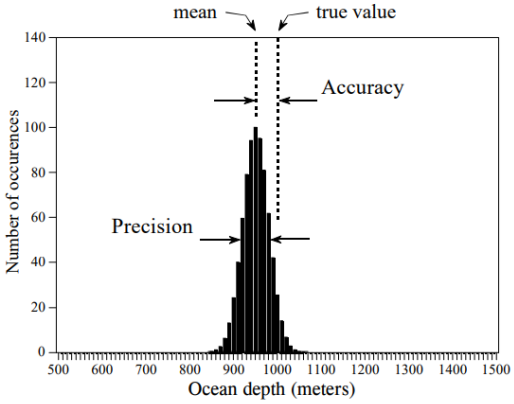


Figure 7 Definitions of accuracy and precision.

Consider a measurement that has good accuracy, but poor precision; the histogram is centered over the true value but is very broad. Although the measurements are correct as a group, each individual reading is a poor measure of the true value. This situation is said to have poor repeatability; measurements taken in succession don't agree well.

3. Conclusion:

In conclusion, probability and statistics is very useful in many fields especially in the field of the signal processing. Its main concepts are used in signal processing such as each signal has mean, standard deviation and can be described in histogram, PMF, PDF or Normal distribution curve. Probability is also useful in generation of Digital noise which is a main step in Digital signal processing. The probability is also useful in determining the accuracy and precision of a signal and then deciding which type of error happened to the data.

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